

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

Title	Employee Attrition Prediction Using Workforce Indicators																																																																																																									
Problem Statement	A company wants to analyze workforce behavior and predict employee attrition. Create a DataFrame containing OvertimeHours, JobSatisfactionScore, SalaryLevel, TrainingHours, WorkLifeBalance, YearsAtCompany, PromotionCount, and AttritionStatus. Create Log_OvertimeHours, categorize JobSatisfactionScore as Disengaged, Satisfied, or Highly Engaged, generate a correlation matrix, create a treemap visualization of employee categories, and analyze workforce trends to identify practical retention insights.																																																																																																									
Objectives	• To create an employee workforce DataFrame from a predefined Python dictionary. • To perform logarithmic transformation on employee overtime hours. • To segment employees using job satisfaction scores. • To encode attrition status for numerical analysis. • To generate and interpret a correlation matrix of workforce indicators. • To analyze attrition rates across employee engagement categories. • To identify employees with high attrition risk using workforce conditions. • To create a treemap visualization of employee categories. • To derive retention insights related to workload, satisfaction, salary, training, promotion, and work-life balance.																																																																																																									
Dataset URL Link / File	Predefined dictionary dataset used directly inside the Python program: Table A: Overtime, Satisfaction, Salary and Training Indicators <table><tr><th>Emp.</th><th>Overtime Hours</th><th>Job Satisfaction Score</th><th>Salary Level</th><th>Training Hours</th></tr><tr><td>1</td><td>5</td><td>5</td><td>5</td><td>40</td></tr><tr><td>2</td><td>12</td><td>4</td><td>4</td><td>35</td></tr><tr><td>3</td><td>18</td><td>3</td><td>3</td><td>28</td></tr><tr><td>4</td><td>26</td><td>2</td><td>2</td><td>16</td></tr><tr><td>5</td><td>8</td><td>4</td><td>4</td><td>34</td></tr><tr><td>6</td><td>32</td><td>1</td><td>1</td><td>10</td></tr><tr><td>7</td><td>15</td><td>3</td><td>3</td><td>24</td></tr><tr><td>8</td><td>28</td><td>2</td><td>2</td><td>18</td></tr><tr><td>9</td><td>10</td><td>5</td><td>5</td><td>45</td></tr><tr><td>10</td><td>24</td><td>2</td><td>4</td><td>30</td></tr><tr><td>11</td><td>35</td><td>1</td><td>1</td><td>8</td></tr><tr><td>12</td><td>6</td><td>4</td><td>4</td><td>38</td></tr><tr><td>13</td><td>20</td><td>3</td><td>3</td><td>26</td></tr><tr><td>14</td><td>34</td><td>3</td><td>2</td><td>12</td></tr><tr><td>15</td><td>38</td><td>2</td><td>2</td><td>14</td></tr></table> Table B: Work-Life, Tenure, Promotion and Attrition Indicators <table><tr><th>Emp.</th><th>Work-Life Balance</th><th>Years at Company</th><th>Promotion Count</th><th>Attrition Status</th></tr><tr><td>1</td><td>5</td><td>10</td><td>3</td><td>No</td></tr><tr><td>2</td><td>4</td><td>7</td><td>2</td><td>No</td></tr><tr><td>3</td><td>3</td><td>5</td><td>1</td><td>No</td></tr><tr><td>4</td><td>2</td><td>2</td><td>0</td><td>Yes</td></tr></table>	Emp.	Overtime Hours	Job Satisfaction Score	Salary Level	Training Hours	1	5	5	5	40	2	12	4	4	35	3	18	3	3	28	4	26	2	2	16	5	8	4	4	34	6	32	1	1	10	7	15	3	3	24	8	28	2	2	18	9	10	5	5	45	10	24	2	4	30	11	35	1	1	8	12	6	4	4	38	13	20	3	3	26	14	34	3	2	12	15	38	2	2	14	Emp.	Work-Life Balance	Years at Company	Promotion Count	Attrition Status	1	5	10	3	No	2	4	7	2	No	3	3	5	1	No	4	2	2	0	Yes
Emp.	Overtime Hours	Job Satisfaction Score	Salary Level	Training Hours																																																																																																						
1	5	5	5	40																																																																																																						
2	12	4	4	35																																																																																																						
3	18	3	3	28																																																																																																						
4	26	2	2	16																																																																																																						
5	8	4	4	34																																																																																																						
6	32	1	1	10																																																																																																						
7	15	3	3	24																																																																																																						
8	28	2	2	18																																																																																																						
9	10	5	5	45																																																																																																						
10	24	2	4	30																																																																																																						
11	35	1	1	8																																																																																																						
12	6	4	4	38																																																																																																						
13	20	3	3	26																																																																																																						
14	34	3	2	12																																																																																																						
15	38	2	2	14																																																																																																						
Emp.	Work-Life Balance	Years at Company	Promotion Count	Attrition Status																																																																																																						
1	5	10	3	No																																																																																																						
2	4	7	2	No																																																																																																						
3	3	5	1	No																																																																																																						
4	2	2	0	Yes																																																																																																						

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

	5	4	8	2	No	
	6	1	1	0	Yes	
	7	3	4	1	No	
	8	2	3	0	Yes	
	9	5	11	3	No	
	10	3	6	1	No	
	11	1	1	0	Yes	
	12	4	9	2	No	
	13	3	6	1	No	
	14	2	2	0	Yes	
	15	2	2	0	Yes	
	No external CSV file, URL, database, or online dataset is required.					
	Functions Details	Sl. No.	Function	Package	Description of function	
		1	pd.DataFrame()	Pandas	Creates the workforce DataFrame from the predefined dictionary.	
		2	np.log()	NumPy	Calculates the natural logarithm of OvertimeHours.	
		3	pd.cut()	Pandas	Categorizes job satisfaction and attrition risk into defined groups.	
4		map()	Pandas	Converts AttritionStatus values from Yes/No into 1/0.		
5		corr()	Pandas	Generates the correlation matrix for numerical workforce indicators.		
6		groupby()	Pandas	Groups employee records by engagement or attrition category.		
7		agg()	Pandas	Calculates employee count, attrition count, averages, and summaries.		
8		mean()	Pandas	Calculates overall attrition rate and average workforce indicators.		
9		astype()	Pandas	Converts Boolean risk conditions into integer risk points.		
10		value_counts()	Pandas	Counts employees within each attrition-risk category.		
11		Rectangle()	Matplotlib	Creates proportional rectangles for the treemap visualization.		
12		savefig()/show()	Matplotlib	Saves and displays the employee-category treemap.		
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1: Import Pandas, NumPy, Matplotlib, and Rectangle from matplotlib.patches. Step 2: Create the workforce dataset locally in dictionary format. Step 3: Convert the dictionary into a DataFrame and assign employee numbers as the index. Step 4: Create Log_OvertimeHours using the natural logarithm of OvertimeHours. Step 5: Categorize JobSatisfactionScore as Disengaged, Satisfied, or Highly Engaged. Step 6: Encode AttritionStatus as AttritionFlag using No = 0 and Yes = 1.					

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

	Step 7: Select numerical workforce indicators and generate the correlation matrix. Step 8: Calculate the overall attrition rate and category-wise workforce summaries. Step 9: Compare average workforce indicators for employees who stayed and employees who left. Step 10: Create a rule-based risk score from overtime, satisfaction, salary, work-life balance, and promotion conditions. Step 11: Create a black-background treemap showing employee-category size and attrition rate. Step 12: Interpret workforce trends and recommend suitable employee-retention strategies.
Program	<pre> import pandas as pd import numpy as np import matplotlib.pyplot as plt from matplotlib.patches import Rectangle def print_heading(title): print("\n" + "=" * 100) print(title) print("=" * 100) # 1. Workforce dataset stored locally as a Python dictionary. employee_data = { "OvertimeHours": [5, 12, 18, 26, 8, 32, 15, 28, 10, 24, 35, 6, 20, 34, 38], "JobSatisfactionScore": [5, 4, 3, 2, 4, 1, 3, 2, 5, 2, 1, 4, 3, 3, 2], "SalaryLevel": [5, 4, 3, 2, 4, 1, 3, 2, 5, 4, 1, 4, 3, 2, 2], "TrainingHours": [40, 35, 28, 16, 34, 10, 24, 18, 45, 30, 8, 38, 26, 12, 14], "WorkLifeBalance": [5, 4, 3, 2, 4, 1, 3, 2, 5, 3, 1, 4, 3, 2, 2], "YearsAtCompany": [10, 7, 5, 2, 8, 1, 4, 3, 11, 6, 1, 9, 6, 2, 2], "PromotionCount": [3, 2, 1, 0, 2, 0, 1, 0, 3, 1, 0, 2, 1, 0, 0], "AttritionStatus": ["No", "No", "No", "Yes", "No", "Yes", "No", "Yes", "No", "No", "Yes", "No", "No", "Yes", "Yes"] } # 2. Create and display the DataFrame. df = pd.DataFrame(employee_data) df.index = range(1, len(df) + 1) df.index.name = "Employee" print_heading("ORIGINAL WORKFORCE DATAFRAME") print(df.to_string()) print("\nDataFrame Shape:", df.shape) # 3. Feature engineering and workforce segmentation. df["Log_OvertimeHours"] = np.log(df["OvertimeHours"]).round(3) df["EmployeeCategory"] = pd.cut(df["JobSatisfactionScore"], bins=[0, 2, 3, 5], labels=["Disengaged", "Satisfied", "Highly Engaged"], include_lowest=True) </pre>

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

```
df["AttritionFlag"] = df["AttritionStatus"].map({"No": 0, "Yes": 1})

print_heading("FEATURE ENGINEERING AND EMPLOYEE SEGMENTATION")
print(df[[
    "OvertimeHours", "Log_OvertimeHours", "JobSatisfactionScore",
    "EmployeeCategory", "AttritionStatus"
]].to_string())

# 4. Correlation analysis.
numeric_columns = [
    "OvertimeHours", "JobSatisfactionScore", "SalaryLevel",
    "TrainingHours", "WorkLifeBalance", "YearsAtCompany",
    "PromotionCount", "Log_OvertimeHours", "AttritionFlag"
]
correlation_matrix = df[numeric_columns].corr()

print_heading("CORRELATION MATRIX")
print(correlation_matrix.round(3).to_string())

# 5. Workforce trend analysis.
overall_attrition_rate = df["AttritionFlag"].mean() * 100
category_summary = (
    df.groupby("EmployeeCategory", observed=False)
    .agg(
        EmployeeCount=("AttritionFlag", "size"),
        AttritionCount=("AttritionFlag", "sum"),
        AverageOvertime=("OvertimeHours", "mean"),
        AverageSalaryLevel=("SalaryLevel", "mean")
    )
    .reset_index()
)
category_summary["AttritionRatePercent"] = (
    category_summary["AttritionCount"]
    / category_summary["EmployeeCount"] * 100
).round(2)
category_summary["AverageOvertime"] =
category_summary["AverageOvertime"].round(2)
category_summary["AverageSalaryLevel"] =
category_summary["AverageSalaryLevel"].round(2)

status_summary = df.groupby("AttritionStatus")[[
    "OvertimeHours", "JobSatisfactionScore", "SalaryLevel",
    "TrainingHours", "WorkLifeBalance", "YearsAtCompany",
    "PromotionCount"
]].mean().round(2)

print_heading("WORKFORCE TREND ANALYSIS")
print("Overall Attrition Rate:
{:.2f}%".format(overall_attrition_rate))
print("\nEmployee Category Summary")
print(category_summary.to_string(index=False))
print("\nAverage Workforce Indicators by Attrition Status")
print(status_summary.to_string())

# 6. Rule-based attrition risk analysis.
df["RiskScore"] = (
    (df["OvertimeHours"] >= 25).astype(int)
    + (df["JobSatisfactionScore"] <= 2).astype(int)
    + (df["SalaryLevel"] <= 2).astype(int)
    + (df["WorkLifeBalance"] <= 2).astype(int)
    + (df["PromotionCount"] == 0).astype(int)
)
df["AttritionRisk"] = pd.cut(
    df["RiskScore"], bins=[-1, 1, 4, 5], labels=["Low", "Medium",
"High"]
)
```

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

```
high_risk_employees = df[df["AttritionRisk"] == "High"][["OvertimeHours", "JobSatisfactionScore", "SalaryLevel", "WorkLifeBalance", "PromotionCount", "RiskScore", "AttritionRisk", "AttritionStatus"]]

print_heading("ATTRITION RISK ANALYSIS")
print("Risk Category Distribution")
print(df["AttritionRisk"].value_counts().sort_index().to_string())
print("\nHigh-Risk Employees")
print(high_risk_employees.to_string())

# 7. Treemap visualization using proportional Matplotlib rectangles.
total_employees = category_summary["EmployeeCount"].sum()
fig, ax = plt.subplots(figsize=(11, 6))
fig.patch.set_facecolor("black")
ax.set_facecolor("black")
x_position = 0.0
grayscale_colors = ["#2F2F2F", "#555555", "#7A7A7A"]

for index, row in category_summary.iterrows():
    width = row["EmployeeCount"] / total_employees
    ax.add_patch(Rectangle(
        (x_position, 0), width, 1,
        facecolor=grayscale_colors[index], edgecolor="white",
        linewidth=2
    ))
    label = (
        f'{row["EmployeeCategory"]}\n'
        f'Employees: {int(row["EmployeeCount"])}\n'
        f'Attrition Rate: {row["AttritionRatePercent"]:.1f}%'
    )
    ax.text(
        x_position + width / 2, 0.5, label,
        color="white", ha="center", va="center",
        fontsize=12, fontweight="bold"
    )
    x_position += width

ax.set_xlim(0, 1)
ax.set_ylim(0, 1)
ax.axis("off")
ax.set_title(
    "Treemap of Employee Engagement Categories",
    color="white", fontsize=16, fontweight="bold", pad=18
)
plt.tight_layout()
plt.savefig(
    "hr_employee_category_treemap.png",
    dpi=300, facecolor="black", bbox_inches="tight"
)
plt.show()

# 8. Final interpretation.
attrition_correlations = (
    correlation_matrix["AttritionFlag"]
    .drop("AttritionFlag")
    .sort_values(ascending=False)
)

print_heading("FINAL WORKFORCE INSIGHTS")
print("Correlation of Workforce Indicators with Attrition")
print(attrition_correlations.round(3).to_string())
print("\nInterpretation:")
print(
    "1. Higher overtime is associated with greater attrition risk.\n"
```

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

```
"2. Lower job satisfaction, salary level, work-life balance,
training,\n"
" tenure, and promotion opportunities are associated with
attrition.\n"
"3. Disengaged employees show the highest attrition rate.\n"
"4. Retention strategies should prioritize workload control,
employee\n"
" engagement, career growth, training, and work-life balance."
)
```

**Output
Snapshots
With
Explanation/
Observations**

Screenshot 1: Original Workforce DataFrame

ORIGINAL WORKFORCE DATAFRAME									
Employee	OvertimeHours	JobSatisfactionScore	SalaryLevel	TrainingHours	WorkLifeBalance	YearsAtCompany	PromotionCount	AttritionStatus	
1	5	5	5	40	5	10	3	No	
2	12	4	4	30	4	7	2	No	
3	18	3	3	20	3	5	1	No	
4	25	2	2	10	2	2	0	Yes	
5	8	4	4	30	4	8	2	No	
6	32	1	1	10	1	1	0	Yes	
7	15	3	3	20	3	4	1	No	
8	28	2	2	10	2	3	0	Yes	
9	10	3	3	30	3	11	3	No	
10	24	2	4	20	3	6	1	No	
11	35	1	1	0	1	0	0	Yes	
12	6	4	4	30	4	9	2	No	
13	20	3	3	20	3	6	1	No	
14	34	2	2	12	2	2	0	Yes	
15	38	2	2	14	2	2	0	Yes	

Figure 1.1 - Original Workforce DataFrame

Explanation/Observation: The locally defined dictionary was converted into a DataFrame containing 15 employee records and all eight required workforce indicators.

Screenshot 2: Feature Engineering and Employee Segmentation

FEATURE ENGINEERING AND EMPLOYEE SEGMENTATION					
Employee	OvertimeHours	Log_OvertimeHours	JobSatisfactionScore	EmployeeCategory	AttritionStatus
1	5	1.609	5	Highly Engaged	No
2	12	2.485	4	Highly Engaged	No
3	18	2.890	3	Satisfied	No
4	25	3.228	2	Disengaged	Yes
5	8	2.079	4	Highly Engaged	No
6	32	3.466	1	Disengaged	Yes
7	15	2.708	3	Satisfied	No
8	28	3.332	2	Disengaged	Yes
9	10	2.303	5	Highly Engaged	No
10	24	3.178	2	Disengaged	No
11	35	3.555	1	Disengaged	Yes
12	6	1.792	4	Highly Engaged	No
13	20	2.996	3	Satisfied	No
14	34	3.528	3	Satisfied	Yes
15	38	3.638	2	Disengaged	Yes

Figure 1.2 - Logarithmic Overtime Feature and Employee Categories

Explanation/Observation: Log_OvertimeHours was generated using np.log(). JobSatisfactionScore was categorized as Disengaged, Satisfied, or Highly Engaged, and AttritionStatus was retained for comparison.

Screenshot 3: Correlation Matrix

CORRELATION MATRIX									
	OvertimeHours	JobSatisfactionScore	SalaryLevel	TrainingHours	WorkLifeBalance	YearsAtCompany	PromotionCount	Log_OvertimeHours	AttritionFlag
OvertimeHours	1.000	-0.864	-0.877	-0.922	-0.914	-0.906	-0.915	0.868	0.863
JobSatisfactionScore	-0.864	1.000	0.890	0.971	0.906	0.927	-0.874	-0.726	-0.874
SalaryLevel	-0.877	0.890	1.000	0.977	0.880	0.945	-0.862	-0.851	-0.861
TrainingHours	-0.922	0.971	0.977	1.000	0.976	0.969	-0.898	-0.875	-0.875
WorkLifeBalance	-0.914	0.906	0.880	0.976	1.000	0.973	-0.868	-0.863	-0.858
YearsAtCompany	-0.906	0.927	0.945	0.969	0.973	1.000	-0.892	-0.886	-0.886
PromotionCount	-0.915	-0.874	-0.862	-0.898	-0.868	-0.892	1.000	-0.823	-0.820
Log_OvertimeHours	0.868	-0.726	-0.851	-0.875	-0.863	-0.886	-0.823	1.000	0.773
AttritionFlag	0.863	-0.874	-0.861	-0.875	-0.858	-0.886	-0.820	0.773	1.000

Figure 1.3 - Correlation Matrix of Workforce Indicators

Explanation/Observation: OvertimeHours has a strong positive relationship with attrition, while satisfaction, salary, training, work-life balance, tenure, and promotion indicators show negative relationships with attrition.

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

Screenshot 4: Workforce Trend Analysis

WORKFORCE TREND ANALYSIS									
Overall Attrition Rate: 40.00%									
Employee Category Summary									
EmployeeCategory	EmployeeCount	AttritionCount	AverageOvertime	AverageSalaryLevel	AttritionRatePercent				
Disengaged	6	5	30.50	2.00	83.33				
Satisfied	4	1	21.75	2.75	25.00				
Highly Engaged	5	0	8.20	4.40	0.00				
Average Workforce Indicators by Attrition Status									
AttritionStatus	OvertimeHours	JobSatisfactionScore	SalaryLevel	TrainingHours	WorkLifeBalance	YearsAtCompany	PromotionCount		
No	13.11	3.67	3.88	33.33	3.78	7.33	1.78		
Yes	32.17	1.83	1.67	13.00	1.67	1.83	0.00		

Figure 1.4 - Attrition Rate and Workforce Category Summary

Explanation/Observation: The overall attrition rate is 40%. Disengaged employees have the highest attrition rate and the highest average overtime while highly engaged employees show no attrition in the sample.

Screenshot 5: Attrition Risk Analysis

ATTRITION RISK ANALYSIS									
Risk Category Distribution									
AttritionRisk									
Low 3									
Medium 1									
High 5									
High-Risk Employees									
Employee	OvertimeHours	JobSatisfactionScore	SalaryLevel	WorkLifeBalance	PromotionCount	RiskScore	AttritionRisk	AttritionStatus	
4	26	2	2	2	0	5	High	Yes	
6	32	1	1	1	0	5	High	Yes	
8	28	2	2	2	0	5	High	Yes	
13	35	1	1	1	0	5	High	Yes	
15	38	2	2	2	0	5	High	Yes	

Figure 1.5 - Rule-Based Attrition Risk Distribution

Explanation/Observation: Employees were assigned risk points based on overtime, satisfaction, salary, work-life balance, and promotion conditions. Five employees were identified as high-risk, and all five have an observed attrition status of Yes.

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

Screenshot 6: Employee Category Treemap

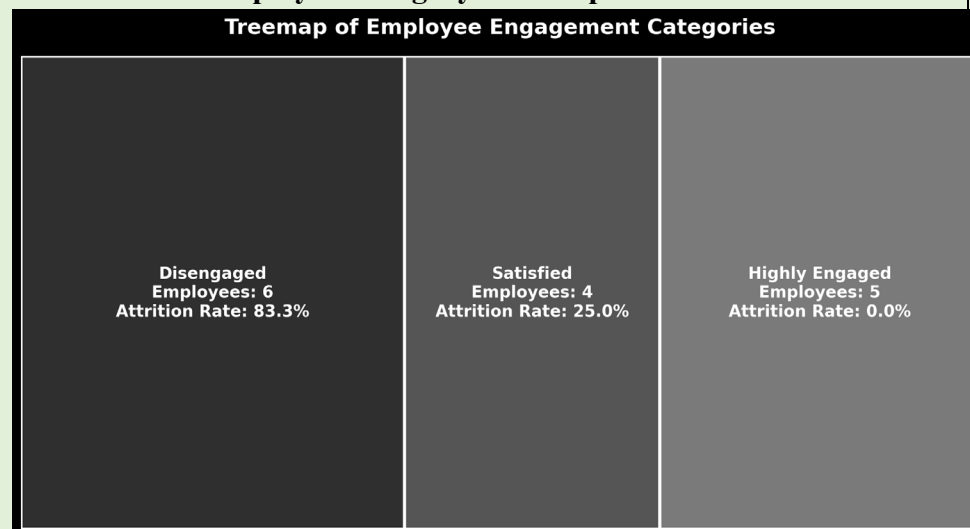


Figure 1.6 - Treemap of Employee Engagement Categories

Explanation/Observation: The treemap displays the relative size of each employee category. Each rectangle also reports the number of employees and the attrition rate for that category.

Screenshot 7: Final Workforce Insights

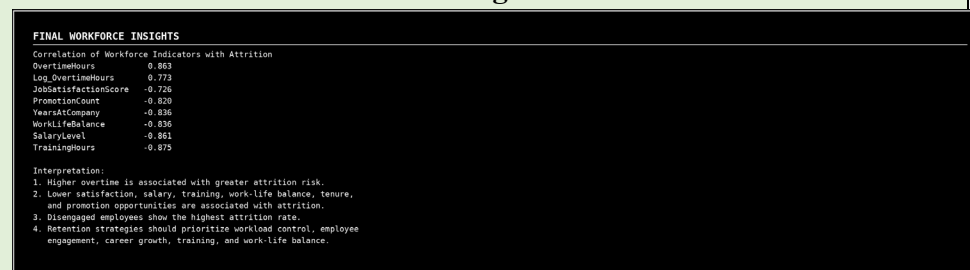


Figure 1.7 - Correlation-Based Workforce Interpretation

Explanation/Observation: The final analysis indicates that overtime is the strongest positive attrition indicator. Retention efforts should focus on workload control, engagement, compensation, training, career growth, and work-life balance.

Conclusions

The Human Resource Analytics exercise was completed successfully using Pandas, NumPy, and Matplotlib. A workforce DataFrame was created from a predefined Python dictionary, Log_OvertimeHours was generated, and employees were segmented using job satisfaction scores. The correlation matrix showed that higher overtime was positively associated with attrition, whereas job satisfaction, salary level, training, work-life balance, tenure, and promotion opportunities were negatively associated with attrition. The treemap highlighted the distribution of engagement categories, and the trend analysis showed that disengaged employees had the highest attrition rate. The rule-based risk analysis identified employees requiring immediate attention. These findings support retention strategies focused on workload reduction, employee

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 30/07/2026

Week No.: 4

Program S. No. 8

	engagement, fair compensation, career progression, training opportunities, and improved work-life balance.
References	1. Pandas documentation for DataFrame creation, categorization, grouping, aggregation, mapping, and correlation analysis. 2. NumPy documentation for logarithmic transformation. 3. Matplotlib documentation for figures, rectangles, text, and image export. 4. Python program supplied with this exercise (https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%204/EXERCISE_PROGRAM.py) 5. Programming for Data Science (MACSE502) class lecture notes (https://drive.google.com/file/d/1PqsD1tC9aCgGFZo7QIJblBulpIjg4UBs/view)

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore